CP1.6: Matrices

1. Matrix Basics

A **matrix** is a rectangular array of numbers. An $m \times n$ matrix has m rows and n columns. The entry in row i, column j is denoted a_{ij}.

Addition and Scalar Multiplication

Matrices of the same size can be added by adding corresponding entries. Scalar multiplication multiplies every entry by the scalar. These operations are entry-wise.

Example:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + 2\begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1+0 & 2+2 \\ 3-2 & 4+4 \end{pmatrix} = \begin{pmatrix} 1 & 4 \\ 1 & 8 \end{pmatrix}$$

Matrix Multiplication

To multiply matrices A (of size $m \times n$) by B (of size $n \times p$), the result AB has size $m \times p$. The (i, j) entry of AB is the dot product of row i of A with column j of B:

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

Note: $AB \neq BA$ in general — matrix multiplication is *not commutative*.

Example (2×2):

$$\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 2(1)+3(-1) & 2(0)+3(2) \\ 1(1)+4(-1) & 1(0)+4(2) \end{pmatrix} = \begin{pmatrix} -1 & 6 \\ -3 & 8 \end{pmatrix}$$

Example (3×3):

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 1 & -1 \\ 0 & 4 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & -1 & 3 \end{pmatrix}$$

$(AB)_{11} = 1(1)+2(2)+0(0) = 5$; $(AB)_{12} = 1(0)+2(1)+0(-1) = 2$; etc.

2. Determinant of a 2×2 Matrix

For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$:

$$\det(A) = |A| = ad - bc$$

Geometric meaning: $|\det(A)|$ is the scale factor by which the matrix A changes areas. If $\det(A) > 0$, orientation is preserved; if $\det(A) < 0$, orientation is reversed (reflected).

A matrix is **singular** if $\det(A) = 0$. A singular matrix has no inverse. Geometrically, it maps all points onto a line (or the origin), so areas become zero.

3. Determinant of a 3×3 Matrix

For a 3×3 matrix, expand along the first row (or any row/column):

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg)$$

The signs alternate: $+, -, +$ along a row or column.

Cofactor Expansion

The **cofactor** C_{ij} of entry a_{ij} is $(-1)^{i+j} M_{ij}$, where M_{ij} is the **minor** (determinant of the matrix with row i and column j deleted).

The sign pattern for a 3×3 cofactor matrix is:

$$\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

Worked 3×3 Determinant: Find $\det(A)$ where $A = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 4 & -2 \\ 1 & 0 & 5 \end{pmatrix}$.

Expand along row 1:

$$\det(A) = 2\det\begin{pmatrix} 4 & -2 \\ 0 & 5 \end{pmatrix} - (-1)\det\begin{pmatrix} 0 & -2 \\ 1 & 5 \end{pmatrix} + 3\det\begin{pmatrix} 0 & 4 \\ 1 & 0 \end{pmatrix}$$

$$= 2(4 \cdot 5 - (-2)(0)) + 1(0 \cdot 5 - (-2)(1)) + 3(0 \cdot 0 - 4 \cdot 1)$$

$$= 2(20) + 1(2) + 3(-4)$$

$$= 40 + 2 - 12 = 30$$

Exam tip: When expanding a 3×3 determinant, choose the row or column with the most zeros to minimise arithmetic. Expanding along a row/column with two zeros means only one minor to compute!

4. Transpose of a Matrix

The **transpose** A^T of a matrix A is obtained by swapping rows and columns: $(A^T)_{ij} = A_{ji}$.

Key properties:

- $(A^T)^T = A$
- $(AB)^T = B^T A^T$
- $\det(A^T) = \det(A)$

5. Inverse of a 2×2 Matrix

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $\det(A) = ad - bc \neq 0$:

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Rule: swap a and d, negate b and c, then divide by the determinant.

Example: Find A^{-1} for $A = \begin{pmatrix} 3 & 2 \\ 5 & 4 \end{pmatrix}$.

$\det(A) = 12 - 10 = 2$

$$A^{-1} = \frac{1}{2} \begin{pmatrix} 4 & -2 \\ -5 & 3 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -5/2 & 3/2 \end{pmatrix}$$

6. Inverse of a 3×3 Matrix

The inverse of a 3×3 matrix uses the cofactor (adjugate) method:

$$A^{-1} = \frac{1}{\det(A)}\text{adj}(A) = \frac{1}{\det(A)}C^T$$

where C is the **cofactor matrix** and C^T is its transpose (the **adjugate**).

Steps:

1. Compute $\det(A)$. If zero, A is singular (no inverse).
2. Find all 9 cofactors $C_{ij} = (-1)^{i+j}M_{ij}$.
3. Form the cofactor matrix C.
4. Transpose to get adj$(A) = C^T$.
5. Divide by $\det(A)$.

Example: Find A^{-1} for $A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 2 & 0 & -1 \end{pmatrix}$.

Step 1: $\det(A)$
Expanding along row 1:
$\det(A) = 1(3(-1) - 1(0)) - 2(0(-1) - 1(2)) + 0(\ldots) = 1(-3) - 2(-2) = -3 + 4 = 1$

Step 2: Cofactors
$C_{11} = +(3(-1) - 1 \cdot 0) = -3$
$C_{12} = -(0(-1) - 1 \cdot 2) = -(-2) = 2$
$C_{13} = +(0 \cdot 0 - 3 \cdot 2) = -6$
$C_{21} = -(2(-1) - 0 \cdot 0) = -(-2) = 2$
$C_{22} = +(1(-1) - 0 \cdot 2) = -1$
$C_{23} = -(1 \cdot 0 - 2 \cdot 2) = -(-4) = 4$
$C_{31} = +(2 \cdot 1 - 0 \cdot 3) = 2$
$C_{32} = -(1 \cdot 1 - 0 \cdot 0) = -1$
$C_{33} = +(1 \cdot 3 - 2 \cdot 0) = 3$

Step 3–5: adj(A) = C^T, then divide by det

$$A^{-1} = \frac{1}{1}\begin{pmatrix} -3 & 2 & 2 \\ 2 & -1 & -1 \\ -6 & 4 & 3 \end{pmatrix}$$

Verify: $AA^{-1} = I$.

7. Solving Simultaneous Equations

A system of equations $A\mathbf{x} = \mathbf{b}$ (where A is the coefficient matrix) has the solution:

$$\mathbf{x} = A^{-1}\mathbf{b} \quad (\text{provided } A \text{ is non-singular})$$

Example: Solve the system: $x + 2y = 5,\ 3x - y = 4$.

Matrix form: $\begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$

$\det = -1 - 6 = -7$, so $A^{-1} = \frac{1}{-7}\begin{pmatrix} -1 & -2 \\ -3 & 1 \end{pmatrix}$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{-7}\begin{pmatrix} -1 & -2 \\ -3 & 1 \end{pmatrix}\begin{pmatrix} 5 \\ 4 \end{pmatrix} = \frac{1}{-7}\begin{pmatrix} -13 \\ -11 \end{pmatrix} = \begin{pmatrix} 13/7 \\ 11/7 \end{pmatrix}$$

8. Key Properties and Rules

Property	Formula	Notes
Commutativity	$AB \neq BA$ in general	Order matters in multiplication
Associativity	$A(BC) = (AB)C$	Always holds

Inverse product	$(AB)^{-1} = B^{-1}A^{-1}$	Order reverses
Determinant of product	$\det(AB) = \det(A)\det(B)$	Multiplicative
Determinant of inverse	$\det(A^{-1}) = \frac{1}{\det(A)}$	
Transpose of product	$(AB)^T = B^T A^T$	Order reverses
Area scale factor	$\lvert\det(A)\rvert$	By how much A scales areas

Exam tip: When a question says "find the values of k for which the matrix is singular", set $\det = 0$ and solve. This is a very common question type. Also: if asked to "show that a matrix is non-singular", compute the determinant and show it is non-zero.